Supplementary Tables for Rebuttal

Table 1. Test RMSE comparison under input-dependent heteroscedastic noise.

Noise type	Ours	LS	LAD	Huber	Cauchy	Tuke
$\epsilon \sim \mathcal{N}(0, 3X_{i1}+4X_{i2})$	**0.2875**	0.4316	0.3653	0.3628	0.3736	10.17
$\epsilon \sim \mathcal{N}(0, (0.5+\sin^2(X_{i1})+X_{i2}^2)^2)$	**0.2674**	0.5470	0.3783	0.3825	0.3635	10.79

Table 2. Comparison with data-driven loss learning methods on Drug Discovery dataset. (Mean and variance over multiple runs).

Noise Rate	Ours	Barron	TERM	GPLFL	ML3	EvoMAL	Tayl
0.2	**0.90(0.06)**	1.04(0.05)	1.45(0.03)	2.04(0.34)	1.89(0.57)	1.67(0.30)	1.4
0.4	**1.00(0.06)**	1.38(0.05)	1.49(0.04)	1.41(0.52)	2.15(0.74)	1.45(0.28)	1.6
0.8	**1.31(0.04)**	3.11(0.20)	1.57(0.04)	1.72(0.58)	2.02(0.69)	1.67(0.43)	1.6

Table 3. Ablation study on the Drug dataset varying the Meta Size.

Meta Size	Noise=0.2 (RMSE/SE)	Noise=0.4 (RMSE/SE)	Noise=0.8 (RMSE/SE)
64	0.9188 / 0.0641	0.9778 / 0.0580	1.2996 / 0.0487
128	0.9362 / 0.0629	0.9847 / 0.0627	1.3022 / 0.0456
256	0.9069 / 0.0638	0.9991 / 0.0598	1.3719 / 0.0421
384	0.9209 / 0.0649	1.0178 / 0.0585	1.3645 / 0.0444
512	0.9233 / 0.0638	0.9767 / 0.0626	1.3589 / 0.0441
768	0.9188 / 0.0636	1.0234 / 0.0595	1.4274 / 0.0418

Table 4. Performance comparison without requiring any clean meta-data (Ours-DAC-MR).

Method	avg test rmse	se
Ours (Meta Size: 128)	**0.94**	0.06
Ours-DAC-MR *(no clean meta set)*	1.00	0.06

TaylorGLO	1.41	0.53
TERM	1.45	0.03
EvoMAL	1.67	0.30

Table 5. Performance on real-world regression benchmarks (Test RMSE).

Method	Shift15m	Abalone
Ours	**9,987.30**	**2.1683**
TaylorGLO	11,567.09	2.8022
Cauchy	11,919.28	2.5959
EvoMAL	11,471.06	2.3370
GMM	12,114.39	2.7178
LAD	12,476.91	2.4471
Huber	13,313.19	2.7890
Barron	14,296.91	3.0898
GPLFL	11,925.58	2.3680
ML3	24,551.25	4.4900
LS	27,067.82	3.7286

Table 6. Comparison with GMM, Truncated, and Welsch losses on Drug Dataset (Test RMSE).

Noise Rate	Ours	Barron	GMM	Trunc	Welsch
0.2	**0.90(0.06)**	1.04(0.05)	1.02(0.06)	0.99(0.06)	0.98(0.06)
0.4	**1.00(0.06)**	1.38(0.05)	1.23(0.06)	1.05(0.05)	1.38(0.04)
0.8	**1.31(0.04)**	3.11(0.20)	3.02(0.15)	1.86(0.04)	3.11(0.16)

Table 7. Sensitivity to hyperparameter `meta_lr`.

meta_lr	1×10^{-5}	3×10^{-5}	1×10^{-4}	$3\times10^{-4}
Test RMSE	1.462 (0.037)	1.301 (0.046)	1.302 (0.046)	1.303 (0.046)

Table 8. Sensitivity to hyperparameter `num_rff`.

num_rff	256	512	1024	2048	4096
Test RMSE	1.305 (0.044)	1.304 (0.045)	1.302 (0.046)	1.305 (0.046)	1.302 (0.045)

Table 9. Sensitivity to hyperparameter `rff_sigmas`.

rff_sigmas	0.8	1.0	2.0
Test RMSE	1.311 (0.046)	1.310 (0.046)	1.312 (0.045)

Table 10. Training time per Epoch on Drug dataset.

Method	Time (seconds)	Method	Time (seconds)
LS	0.40	Tilting	0.40
LAD	0.40	GMM	0.50
Huber	0.45	Trunc	0.45
Cauchy	0.45	Welsch	0.40
Tukey	0.50	Barron	0.50
Ours	**1.70**		

Table 11. Comparison with tuned fixed losses and adaptive-parameter methods (Drug dataset, noise = 0.2).

Method	Avg Test RMSE (SE)
Ours	**0.94 (0.06)**
Huber ($\xi=0.5$ / $\xi=1.345$)	1.0197 (0.0555) / 1.0436 (0.0554)
Cauchy ($\kappa=0.5$ / $\kappa=1.0$)	1.2856 (0.0144) / 1.2859 (0.0145)
Cauchy ($\kappa=1.345$ / $\kappa=2.0$ / $\kappa=3.0$)	~1.285 (0.014)
Barron	1.04 (0.05)
TERM	1.45 (0.03)

Figure 1. Runtime per Epoch (s) vs. Inner Iteration Steps (K).

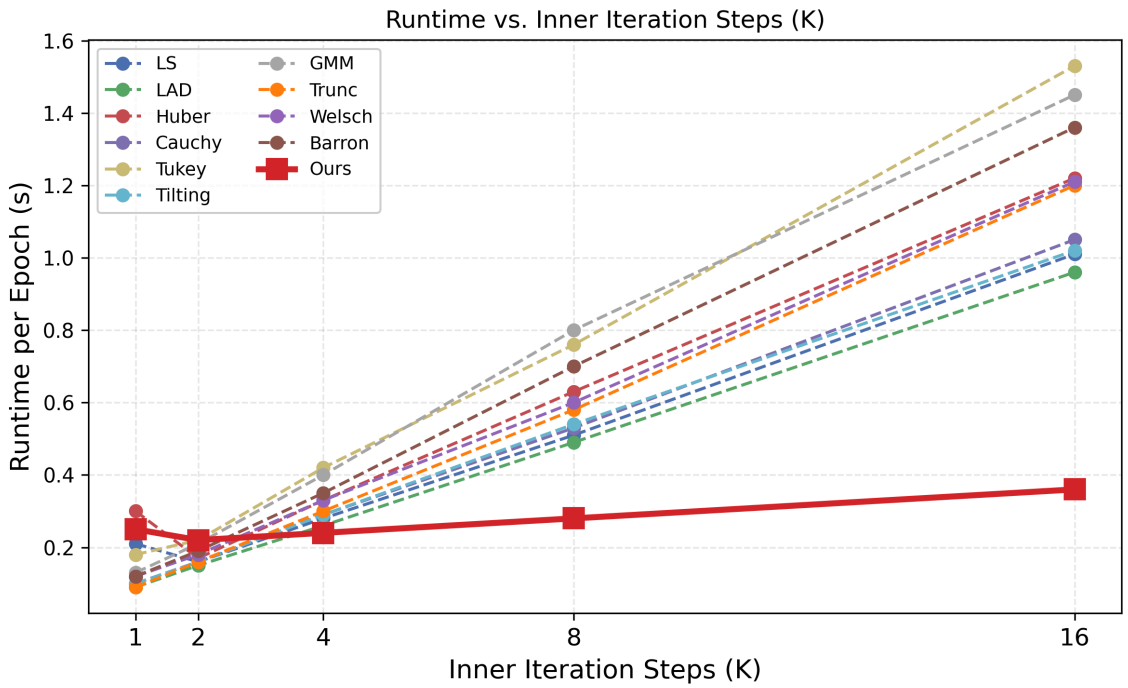
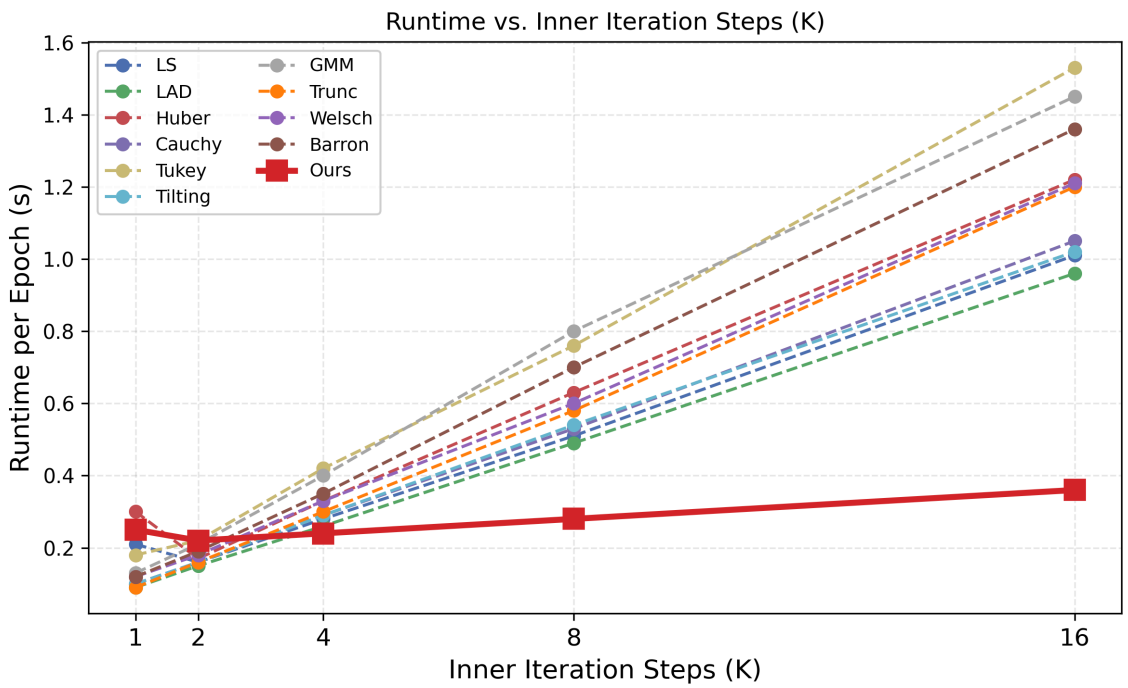
Figure 2. Memory Usage (MB) vs. Inner Iteration Steps (K).